

Technical Blueprint: IMCI Data Pipeline

You have selected the perfect file (WHO IMCI). These are the exact 4 technical steps you need to execute now in your code editor to finish Day 1 of the Hackathon.

The 4 Action Steps

1. Parsing (LlamaParse)



Write a script using `LlamaParse` API to extract the IMCI PDF. Why? Because IMCI is 90% tables (Assess/Classify/Treat). Standard text extraction will destroy this logic. LlamaParse converts these tables into clean Markdown.

2. Chunking (Section-Aware)



Don't split by character count! Write a function to split the Markdown file by Headers (e.g., `# COUGH OR DIFFICULT BREATHING`). Keep chunks around 300-500 tokens. This keeps symptoms and treatments grouped together.

3. Metadata Schema



For every chunk you create, you MUST attach a JSON object containing: `document\_name` (IMCI), `section\_title` (e.g., Diarrhoea), and `page\_number`. This is mandatory for the Citation requirement.

4. Embedding & Chroma DB



Initialize a local `Chroma` client. Pass your chunks through an embedding model (like `OpenAI text-embedding-3` or `sentence-transformers`) and save them to your Chroma collection.

The **Metadata** Payload

- This is the most critical code structure for Day 1.
- When the AI generates an answer on Day 3, it will read this metadata to tell the doctor exactly where the info came from.
- If you skip this step, your RAG system will fail the "Citation" rule.

```
// Example Chunk Object in your code
{
  "id": "imci-chunk-045",
  "text": "If child has stridor in calm state,
classify as SEVERE PNEUMONIA...",
  "metadata": {
    "source": "WHO_IMCI_Handbook.pdf",
    "section": "Assess & Classify Cough",
    "page_number": 14,
    "color_code": "Red (Urgent)"
  }
}
```

Your Next Immediate Action

- **1. Open VS Code:** Create a new Python project folder.
- **2. Install Packages:** Run `pip install llama-parse chromadb sentence-transformers` in your terminal.
- **3. Load the PDF:** Place `9241546441.pdf` in a `/data` folder inside your project.
- **4. Write the Parser:** Write the first script to send this PDF to LlamaParse and save the output as `imci\_parsed.md`.

```
ow.adngin && window.adngin.adnginLoaderReady) {  
  number(w3_getStyleValue(document.getElementById("main"), "height"),  
  (document.getElementById("adngin-mid_content-0")) {  
    adngin.queue.push(function(){ adngin.cmd.startAuction([  
      {placement: "adngin-sidebar_sticky-0", adUnit: "sidebar_sticky"},  
      {placement: "adngin-mid_content-0", adUnit: "mid_content"}  
    ]));  
  });  
} else {  
  adngin.queue.push(function(){ adngin.cmd.startAuction([  
    {placement: "adngin-sidebar_sticky-0", adUnit: "sidebar_sticky"},  
    {placement: "adngin-mid_content-0", adUnit: "mid_content"}  
  ]));  
});  
  
e {  
  (document.getElementById("adngin-mid_content-0")) {  
    adngin.queue.push(function(){ adngin.cmd.startAuction([  
      {placement: "adngin-mid_content-0", adUnit: "mid_content"}  
    ]));  
  });  
}
```

Image Sources

600 × 400

http://googleusercontent.com/image_collection/image_retrieval/3117341918233641364_0

Source: en.idei.club

Image Sources



<https://cdn.builtin.com/cdn-cgi/image/f=auto,fit=cover,w=1200,h=635,q=80/sites/www.builtin.com/files/2025-06/dark-mode-coding.jpg>

Source: [builtin.com](https://www.builtin.com)